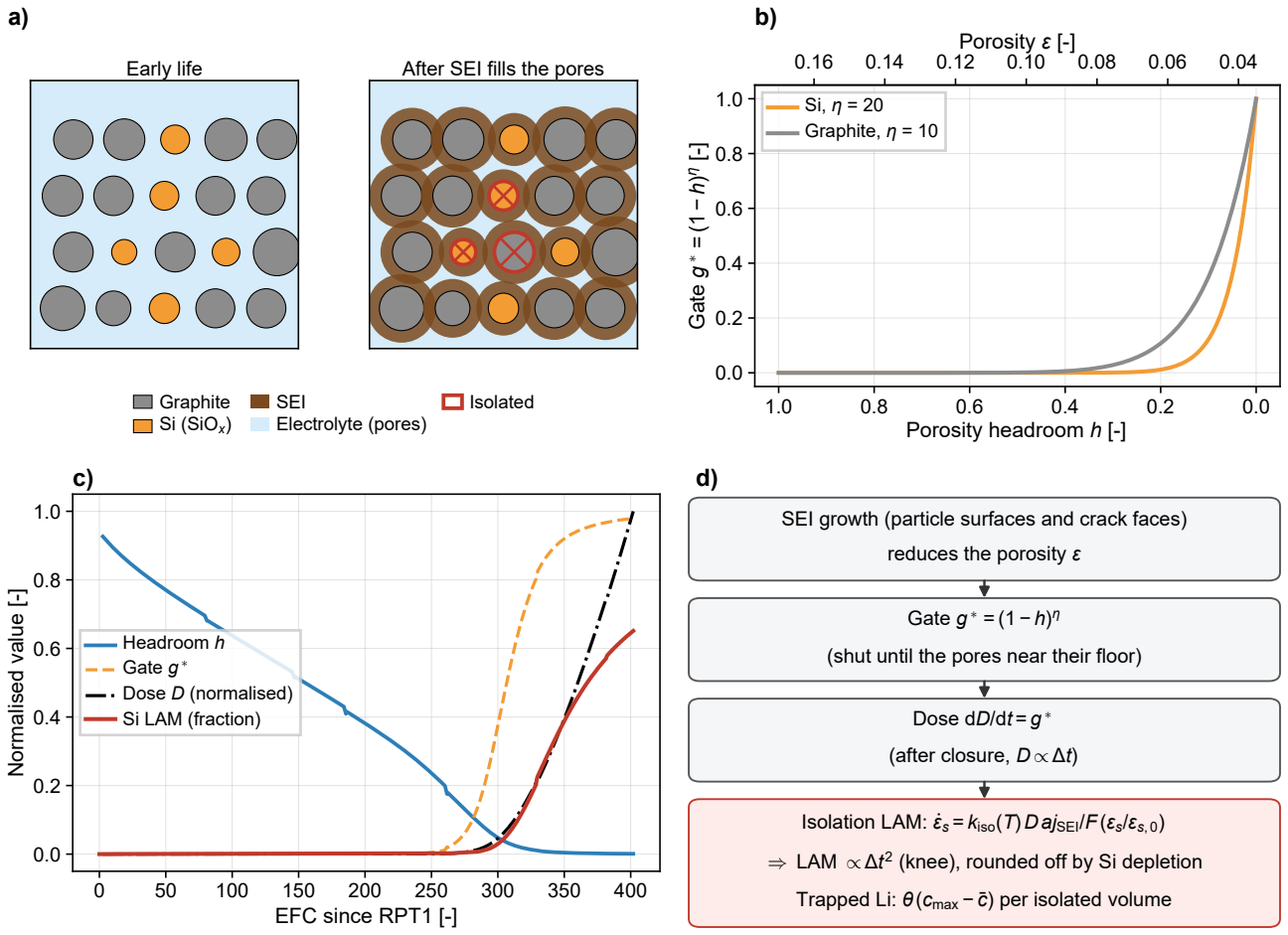

a)
Early life
After SEI fills the pores
Graphite
SEI
Isolated
Si (SiO$_x$)
Electrolyte (pores)
b)
Porosity ε [-]
0.16
0.14
0.12
0.10
0.08
0.06
0.04
Si, η = 20
Graphite, η = 10
Gate $g^* = (1-h)^\eta$ [-]
Porosity headroom h [-]
c)
Normalised value [-]
Headroom h
Gate g^*
Dose D (normalised)
Si LAM (fraction)
EFC since RPT1 [-]
d)
SEI growth (particle surfaces and crack faces)
reduces the porosity ε
Gate $g^* = (1-h)^\eta$
(shut until the pores near their floor)
Dose $\mathrm{d}D/\mathrm{d}t = g^*$
(after closure, $D \propto \Delta t$)
Isolation LAM: $\dot{\varepsilon}_s = k_{\mathrm{iso}}(T)\, D\, a j_{\mathrm{SEI}} / F(\varepsilon_s/\varepsilon_{s,0})$
$\Rightarrow$ LAM $\propto \Delta t^2$ (knee), rounded off by Si depletion
Trapped Li: $\theta(c_{\max} - \bar{c})$ per isolated volume